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Teaching AI One Minute at a Time: Aligning DEFT-AI With the One-Minute Preceptor --Manuscript Draft--

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Teaching AI One Minute at a Time: Aligning DEFT-AI With the One-Minute Preceptor

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The Problem: When Technology Outpaces Teaching

The integration of artificial intelligence (AI) into clinical workflows is accelerating at a pace that surpasses our pedagogical frameworks. AI's rapid adoption in undergraduate and graduate medical education (UME, GME) has been largely norm-driven—fueled by the ubiquity of the technology—rather than guided by structured learning objectives. This shift risks the emergence of a "hidden curriculum" of AI. When residents utilize AI for clinical reasoning and documentation without formal faculty guidance, they risk internalizing an implicit normalization of automation bias.¹ Without deliberate educational interventions, we face a future of "double-checkers"—clinicians who sign off on AI-generated outputs but lack the primary clinical reasoning skills to detect the subtle, high-stakes errors these models can produce.

From Use to Understanding: The Gap in Utilizing and Supervising AI

The educational risks are multidimensional, spanning the spectrum from "deskilling"—the loss of existing diagnostic acumen—to "never-skilling," when learners fail to develop foundational competencies because complex cognitive tasks are prematurely offloaded to digital assistants.¹ This challenge is compounded by a widening "expertise gap": learners may be more technically adept at using AI than supervising faculty. The AAMC's Principles for the Responsible Use of AI in Medical Education, outlines the necessity of preparing our future workforce.² However, a significant implementation gap remains: how can busy preceptors recognize, probe, and validate AI-mediated clinical decisions within real-time clinical workflows constraints?

From Microskills to Machine Skills

Using AI safely demands ongoing oversight and preservation of human clinical judgment,³ which can be achieved by building on the foundational microskills of the One Minute Preceptor (OMP).⁴ By utilizing the DEFT (Diagnosis, Evidence, Feedback, Teaching) framework—a streamlined adaption of the OMP^{5,6}—educators can integrate a necessary meta-cognitive layer specifically designed for the AI era. DEFT-AI does not replace the preceptor’s judgement; rather, it provides a systematic approach to surface automation bias and ensure AI supports rather than weakens clinical reasoning.

Adapting Familiar Teaching Models for an AI-Enabled Era

In this perspective, we expand on prior work integrating DEFT-AI with the OMP, offering a pragmatic way to integrate AI supervision into routine clinical education without adding substantial time or cognitive burden (Table 1). The OMP five-step microskills model provides an especially well-suited scaffold for this integration given its emphasis on efficient, learner-centered teaching in busy clinical environments^{4,5}. DEFT-AI extends these principles to explicitly address learner–AI interactions.⁶

The first DEFT-AI domain (D)—diagnosis, discussion, and discourse—aligns closely with the microskill of “getting a commitment.” Preceptors can ask learners not only to commit to a diagnosis or management plan, but also articulate how AI informed their reasoning, including the prompts used.

The evidence (E) domain maps onto “probing for supporting evidence,” but expands it to include critical appraisal of AI outputs. Encouraging learners to explicitly signal their degree of uncertainty further deepens this step, fostering metacognitive awareness and appropriate calibration of trust in AI systems. This approach reflects emerging recommendations that clinical

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4 supervision of AI should emphasize transparency in reasoning processes and human–AI
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6 interaction patterns.⁷ Learners can be asked how the AI may have generated its conclusions, what
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8 limitations or biases may be present, and whether they independently verified outputs. This
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10 aligns with competency frameworks emphasizing evidence-based evaluation of AI tools,
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12 including their accuracy, safety, and contextual appropriateness.⁸ In practice, this can allow
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14 teaching encounters to address broader digital health skills, such as the ethical and social aspects
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16 of AI.
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22 The feedback (F) component of DEFT-AI corresponds to reinforcing what was done well and
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24 identifying areas for improvement. In addition to traditional clinical feedback, this step explicitly
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26 addresses learners’ use of AI—highlighting effective strategies (e.g., iterative prompting, cross-
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28 verification) and identifying gaps such as overreliance or failure to question outputs. Introducing
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30 concepts such as “cyborg” versus “centaur” may be helpful.⁷ A cyborg approach, characterized
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32 by tight human–AI integration, may be appropriate for high-risk or diagnostically uncertain
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34 scenarios requiring continuous human oversight, whereas a centaur model, involving division of
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36 labor between clinician and AI, may be better suited to lower-risk or well-defined tasks. Such
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38 distinctions help learners develop adaptive strategies for AI use rather than a one-size-fits-all
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40 approach. Unlike the one-minute preceptor’s dedicated “correct mistakes” step, DEFT-AI
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42 embeds error correction across multiple domains, including identifying bias and inappropriate
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44 reliance on AI through reflective discussion and critical appraisal.
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52 The teaching (T) domain aligns with “teaching a general principle,” allowing preceptors to
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54 generalize from the encounter. For example, faculty can offer guidance on when AI use is
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56 appropriate, how to structure effective prompts, or how to recognize unreliable outputs. These
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58 teaching points can be explicitly linked to broader competency frameworks, including domains
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4 such as AI-enhanced clinical encounters, workflow adaptation, and practice-based
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6 improvement.^{8,9}
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10 **From Learners to Systems: Advancing AI Integration in Medical Education**

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14 The integration of AI into clinical education must be intentional. We know that learners will
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16 inevitably develop patterns of AI use through a hidden curriculum, absorbing both effective and
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18 problematic habits without structured reflection or feedback. This is further complicated by a
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20 growing divide, in which “AI-native” learners are more skilled than supervising faculty. Without
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22 a shared framework, we risk normalizing unexamined use rather than fostering critical
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24 engagement.
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29 For educators, this moment calls for a shift toward co-learning models of supervision. Preceptors
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31 do not need to be technical experts in AI to teach responsible use; rather, they can and should
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33 model critical inquiry, uncertainty, and reflective practice alongside learners to create safe
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35 learning environments.¹⁰ While this requires a degree of vulnerability, transparency may
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37 strengthen the learning alliance and reinforce a culture of intellectual humility. Framing AI
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39 discussions within established teaching structures, such as the OMP, allows faculty to maintain
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41 educational authority while adapting to evolving clinical realities.
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48 Beyond the core DEFT-AI and OMP steps, preceptors may recommend direct supervision when
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50 learners are using AI in high-stakes contexts or demonstrate limited critical appraisal skills, and
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52 more indirect supervision as competence develops. This graduated approach aligns with existing
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54 models of clinical supervision and reflects calls for structured oversight of AI integration in
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56 clinical training.⁷
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4 Importantly, DEFT-AI, like OMP, should not be viewed as a rigid checklist to complete for
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6 every patient encounter. The goal is efficient, targeted teaching within busy clinical workflows.
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8 Preceptors may selectively emphasize different DEFT-AI elements depending on the clinical
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10 scenario, learner needs, and degree of AI involvement. For example, faculty may focus on a
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12 different DEFT-AI component with each case rather than addressing every step for every patient.
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14 This flexible approach helps maintain attention to both patient care and thoughtful AI
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16 supervision without substantially increasing time burden.
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22 At the program level, AI integration highlights the growing importance of three interconnected
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24 competencies: independent clinical reasoning, critical appraisal of AI, and effective human–AI
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26 collaboration. Although AI has the potential to create more space for humanistic, patient-
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28 centered care, this should not be assumed; it depends on whether learners are taught to use these
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30 tools thoughtfully, safely, and in alignment with clinical goals.
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35 Taken together, aligning DEFT-AI with the one-minute preceptor model embeds AI literacy into
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37 familiar teaching workflows while advancing competencies identified across multiple
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39 frameworks in medical education.¹¹ This integration offers an example of a pedagogy-first
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41 approach to medical education highlighting how educational objectives guide the integration of
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43 technology, rather than the reverse.
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Table 1. Aligning the DEFT-AI Framework With the One-Minute Preceptor

DEFT-AI Domain	One Minute Preceptor Microskill	Example Teaching Prompt
Diagnosis, Definition, Discourse	Get a Commitment	“What prompted you to use AI?” “How did AI help with your differential diagnosis?” “What prompts did you use for OpenEvidence®?”
Evidence	Probe for supporting evidence	“Tell me about how sure you were of your possible diagnoses when you interacted with AI?” “What limitations do you see in the AI output?” “What information might AI have missed?”
Feedback	Reinforce what was right (Correcting Mistakes) or Identify areas for Improvement)	“Tell me why you did not verify whether the references were correct?” “Do you think your decision-making in the case was high or low stakes for the patient? And how did your use of AI match that?” “Looking back, what would you do differently next time?”
Teaching	Teach General Principles	“When writing a prompt, here are some key considerations....” “In this case, the unusual journal name was a clue that this might have been a hallucination.” “What general lesson can we take from this case about using AI?”

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To the Editor of The Journal of Graduate Medical Education,

I am the corresponding author for the submission entitled, “Teaching AI One Minute at a Time: Aligning DEFT-AI With the One-Minute Preceptor,” submitted as a JGME Perspectives piece. The authors of this paper are Jamie Geringer, DO, MHPE, FACP and Paul Hemmer, MD, MPH, MACP.

Artificial intelligence (AI) tools are increasingly being used in the clinical environments, often without explicit supervision or guidance. As trainees adopt AI in patient care and clinical reasoning, graduate medical education faces the challenge of ensuring that these tools support rather than undermine the development of clinical judgment. While emerging frameworks have described competencies for AI use and approaches to supervising learner–AI interactions, many faculty lack practical strategies for integrating these concepts into busy, time-constrained clinical teaching environments.

In this Perspective, we explore how the One-Minute Preceptor model aligns with the DEFT-AI framework. By mapping DEFT-AI concepts onto existing microskills, we offer practical strategies for supervising learner use of AI while preserving efficiency, learner-centered development and supervision. We also discuss implications for learners, educators, and training programs as AI becomes increasingly embedded in clinical practice.

We believe this manuscript will be of interest to JGME readers because it addresses a timely challenge facing clinical educators and offers a feasible, pedagogy-driven approach to AI integration in graduate medical education. We confirm that this manuscript is original, has not been published, and is not under consideration elsewhere. All authors participated in the preparation of the manuscript and approved its submission. There are no conflicts of interest to disclose.

We look forward to the opportunity to contribute to The Journal of Graduate Medical Education. Please feel free to contact me at jamie.geringer@usuhs.edu or 914 672 5412 with any questions.

Respectfully,

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